

ACCESS KNOB: Parametric Design of 3D-Printable Accessible Synthesizer Controls for Users with Reduced Hand Strength

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Synthesizers and electronic musical instruments are popular tools for creating contemporary music, offering versatile expressions of music compositions that easily transfer to electronic media platforms. While these tools increase the accessibility of music production to broader populations, particularly by reducing barriers for recording and distributing music while also at times lowering the overhead towards instrumental proficiency, the physical design of the instruments present significant accessibility barriers for users with motor impairments affecting grip strength and fine motor control. These barriers unnecessarily deter musicians bearing these impairments from creating music. To address this, the ACCESS KNOB, an open-source parametric web tool for generating customizable 3D-printable knobs, will make it possible to create replacement knobs for synthesizers and electronic musical instruments that can be operated with reduced hand strength and sensation.

CCS Concepts: • **Human-centered computing** → **Accessibility technologies**; *Accessibility design and evaluation methods*; • **Applied computing** → **Sound and music computing**.

Additional Key Words and Phrases: accessibility, assistive technology, parametric design, 3D printing, synthesizers, motor impairments, universal design, music technology

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1 Introduction

Musical expression through synthesizers and electronic instruments requires precise tactile control of numerous parameters, such as knobs, faders, and buttons. For users with motor impairment, reduced grip strength, or diminished hand sensation, these standard interface controls can present insurmountable barriers to the usability of these tools that limit full access to music expression. The typical synthesizer knob—small in diameter, smooth or minimally textured, and requiring precision grip and fine motor control—excludes a significant population of potential users. [1]

This work was commissioned by a retired IT professional and avid synthesizer collector who, due to dialysis-dependent kidney failure, had progressively lost hand strength and sensation. This disability grew to the point where his wall of rare synthesizers, accumulated over decades, had become largely unplayable. Workarounds failed to produce the nuanced, expressive control integral to synthesizer performance.

Although many people will experience decreased fine motor skills and grip strength decline through age or injury, standard synthesizers and electronic instruments favor smaller knob designs with no mainstream alternatives. [2]

ACCESS KNOB leverages the increased availability of 3D printers and laser cutters [3] to allow musicians to generate customized 3D-printable replacement knobs using a parametric web-based tool. Building on the principles of Universal

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Music Design (UMD) ([4], [5]), ACCESS KNOB addresses motor accessibility while maintaining applicability for all users. A video demo of this platform is available [here](#).

We contribute: (1) documentation of a parametric design system for accessible synthesizer controls informed by lived disability experience; (2) an engineering evaluation of the parametric design space, physical print tolerances, and mechanical torque advantages; (3) discussion of extensibility to other instrument types and assistive device domains; and (4) reflection on open-source distribution as an accessibility strategy [6].

2 Related Work

Research in accessibility-focused Digital Musical Instruments (DMIs) has historically prioritized eye-tracking, occupational therapy frameworks, rapid prototyping of bespoke instruments, or entirely novel interfaces to support musicians with diverse motor and cognitive impairments [7–10, 15]. However, these approaches focus on creating new instruments rather than retrofitting existing commercial hardware, leaving classic physical synthesizers largely inaccessible. Parametric and generative design frameworks address this by enabling personalized, custom-fit physical adaptations, such as non-invasive mappable gesture sensing [16] and custom 3D-printed orthopedic grips or pneumatic controls with variable activation forces [11, 12, 14]. Generative systems further democratize fabrication by exploring design spaces algorithmically, removing manual CAD editing bottlenecks [13]. ACCESS KNOB builds on this by utilizing a mutation engine to generate customizable variant batches, ensuring direct user agency in selecting ergonomic shapes.

ACCESS KNOB integrates and promotes UMD’s ethos of sonic agency [17] to increase accessibility of music technology that historically excluded DHH musicians and musicians with motor disabilities. These frameworks emphasize multimodal, cross-disability design by integrating visual, auditory, and haptic feedback channels [18, 19].

This is particularly relevant for DHH musicians who utilize vibrotactile interfaces for sensory reinforcement [21, 22]. Given that vestibular variations in DHH populations can affect motor coordination [23, 24], our parametric customizer combines tactile shapes, voice/facial inputs, and auditory/haptic feedback to unify motor and sensory accessibility.

3 System Description

ACCESS KNOB is a browser-based parametric design tool built using JavaScript, Three.js for 3D visualization, and a custom geometry engine for procedural knob generation. The interface provides a 3D real-time preview, parameter sliders, export options, and a mutation engine for the exploration of generative designs.

3.1 Parametric Controls

Users adjust the following parameters via sliders and numerical inputs (Table 1).

3.2 Mounting, Mutation, and Export Formats

The customizer supports two mounting strategies: **Swap-In Mode** (directly replacing the original knob) and **Slide-Over Mode** (fitting as a non-destructive sleeve over the existing control).

In addition to 3D-printable meshes, ACCESS KNOB features a 2D vector SVG stacking exporter for subtractive fabrication (e.g., laser-cutting or CNC-milling). The exporter slices the 3D model into 2D layer profiles based on user-specified material sheet thickness and laser kerf width (default 0.1 mm). To facilitate quick and accurate manual assembly, the exporter automatically generates two kerf-compensated alignment rods and corresponding slots through all layers, allowing stacked pieces to be precisely aligned and glued.

Table 1. Primary parametric knob design parameters.

Parameter	Range / Options	Purpose
Shape	Circular, square, hexagonal, triangular, pentagonal, octagonal, star, gear	Shape-coding enables tactile and visual differentiation of control functions, supporting users with low vision through recognizable profiles (Fig. 1) [20].
Outer Diameter	15–50 mm (default: 35 mm)	Larger diameters increase mechanical advantage and reduce the grip force required for rotation.
Height	10–40 mm	Taller knobs increase graspable surface area, supporting palm-dragging and diverse grip strategies.
Taper	0–70% Angle	Provides ergonomic grip variation and improves visual and tactile distinction between controls.
Groove Profile	Square, rounded, V-shaped	Affects grip friction, tactile feel, and rotational control.
Bore Diameter	4–10 mm	Matches the shaft diameter of the target synthesizer. A knob database provides common dimensions for popular synthesizers.
Slot Depth	0–3 mm	Controls D-shaft slot depth for secure rotational locking.
Clearance	0–0.5 mm	Adjusts bore fit tolerance for printer precision and preferred friction fit.
Set Screw	Enabled/Disabled; Top or Side	Provides secure mechanical fastening independent of friction fit.

The tool exports manifold STL meshes, OpenSCAD scripts, 2D SVG layer contours, and JSON/CSV datasets for configuration sharing. This architecture ensures that users with motor impairments retain direct design agency over their own instruments rather than relying on proxies, aligning with the sonic agency framework which asserts the right to shape one’s own expressive tools [25].

3.3 Accessibility and Interface Input Options

A critical consideration in assistive technology design is whether the design tool itself is accessible [8]. A parametric knob customization system that requires fine motor control to operate would perpetuate the very barriers it aims to address. ACCESS KNOB’s browser-based architecture supports platform-level accessibility features without custom client-side installation.

Apple Voice Control and Switch Control: macOS and iOS users can operate ACCESS KNOB entirely through voice commands [28] or camera-based facial gestures and head movements [29]. Voice Control enables users to speak commands to navigate, tap buttons, adjust sliders, and trigger exports. Switch Control with camera tracking allows users to control the interface through head movements (left/right gestures) or facial expressions (e.g., smile, raised eyebrows) mapped to interface actions.

Google Project Gameface: Windows and Android users can employ Google’s open-source Project Gameface system [30, 31], which translates head movements and facial gestures into cursor control and click actions. This system operates across any web-based interface, including ACCESS KNOB’s parametric controls and mutation engine.

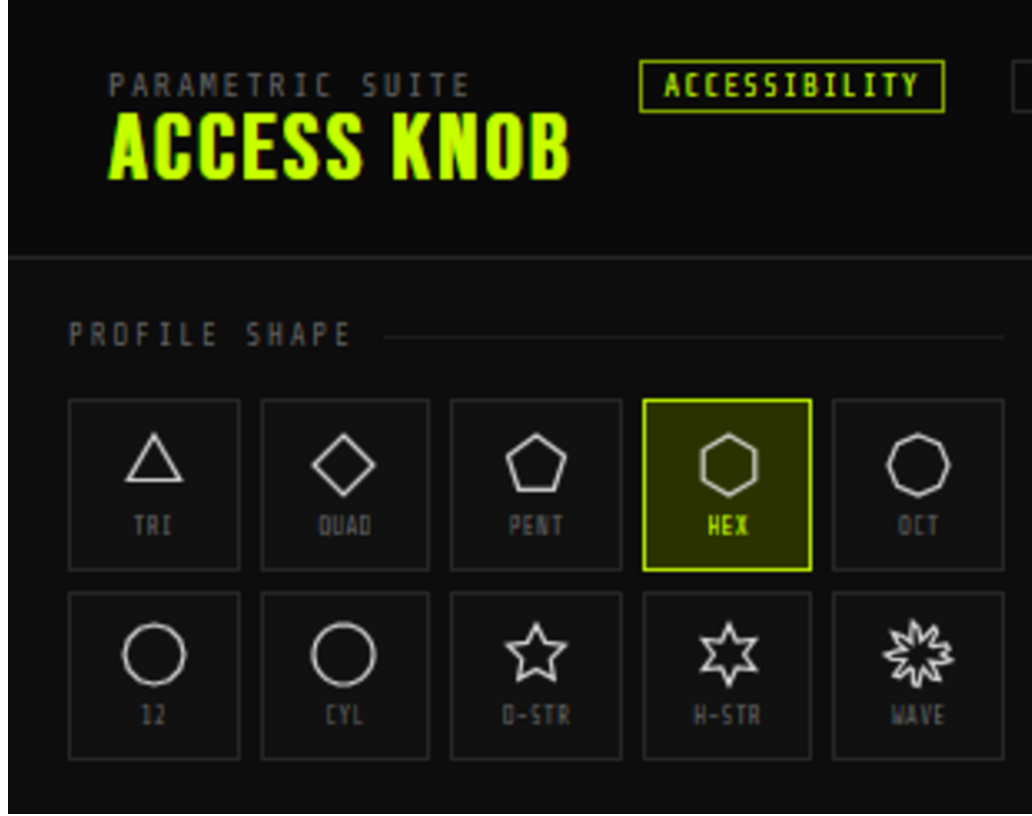


Fig. 1. Profile shape options including triangle, diamond, pentagon, hexagon, octagon, and star configurations for tactile and visual distinction.

Hardware Requirements: Both Apple and Google solutions require only a standard USB webcam, making assistive input accessible at commodity price points. Higher-end implementations (e.g., Apple TrueDepth cameras on recent devices) provide enhanced facial tracking precision but are not required for functional operation.

4 Engineering Evaluation and Design Space Analysis

We evaluate the design space enabled by ACCESS KNOB by analyzing its mechanical advantage, fabrication tolerances, and interface accessibility.

4.1 Mechanical Torque Advantage

To quantify how parametric changes reduce the grip force required to actuate a control, we model the mechanical torque T necessary to rotate a potentiometer or rotary switch:

$$T = F \cdot \left(\frac{D}{2} \right) \implies F = \frac{2T}{D} \quad (1)$$

where F is the applied peripheral finger pinch or grip force, and D is the outer diameter of the knob. This mechanical scaling directly compensates for reduced hand strength without modifying the internal electronic hardware of the

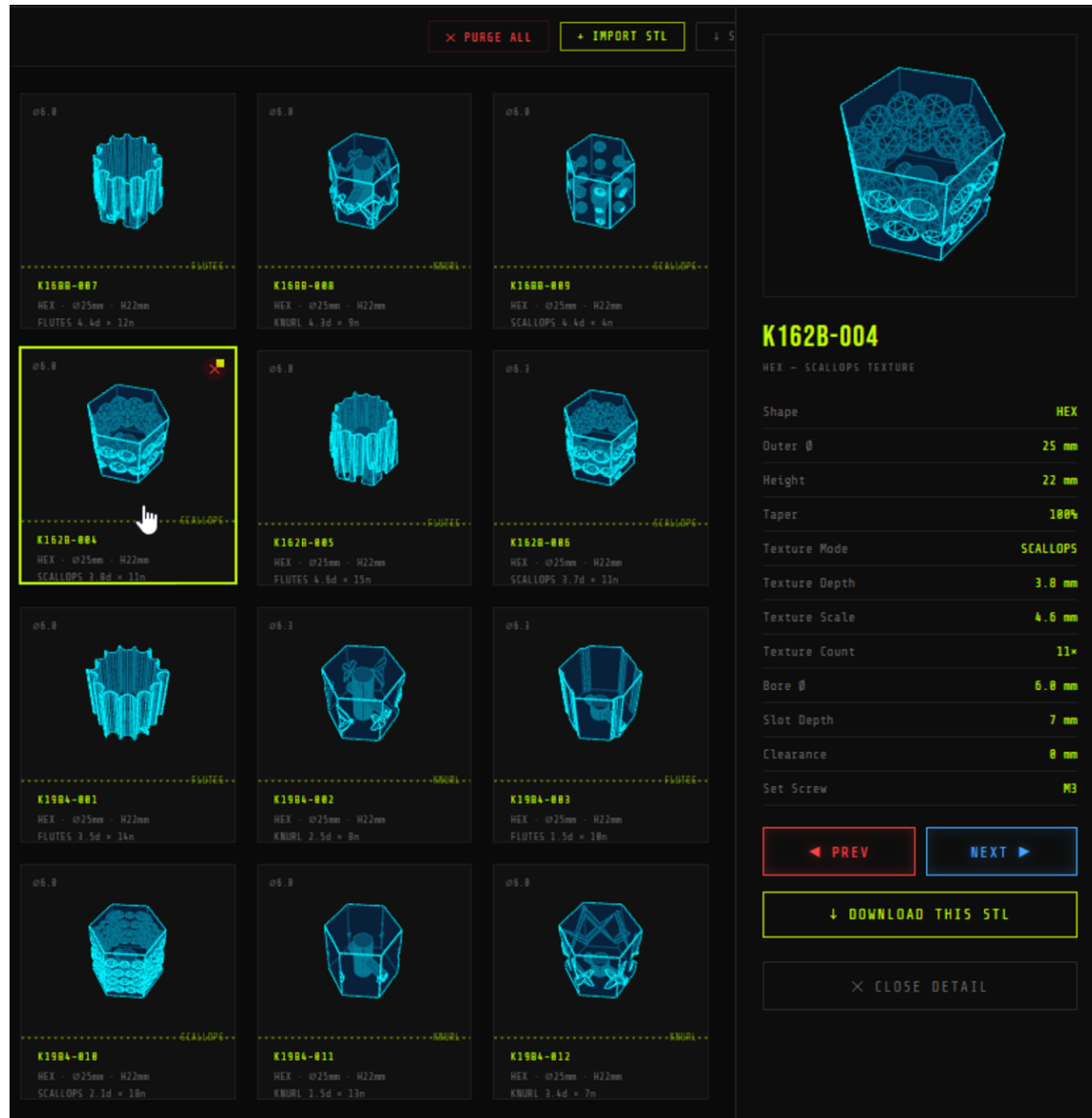


Fig. 2. ACCESS KNOB interface showing a batch of 12 generated knob variants with hexagonal profiles. Parameters are displayed for the selected knob (K793-001) including shape, diameter, height, groove geometry, and bore specifications. The mutation engine enables rapid design exploration.

instrument. Increasing knob height (from standard 12 mm to 25 mm or taller) increases the usable vertical surface area, enabling musicians to employ palm-dragging or side-of-hand shear techniques to actuate the controls.

4.2 Fabrication Tolerances and Materials

4.3 Interface Accessibility Conformance

The browser-based customizer supports keyboard navigation and platform-level hands-free interfaces (Apple Voice/Switch Control, Google Project Gameface) using commodity webcams. For blind or low-vision users, a multimodal interaction loop integrates hidden ‘aria-live’ parameter announcements, oscillator-based parameter sonification (Web Audio API), and short haptic confirmation pulses (‘navigator.vibrate()’).

5 Deployment and Field Study

Our initial deployment targets the individual whose needs catalyzed this project: a retired IT professional (age 64) and avid musician with dialysis-dependent kidney failure resulting in progressive hand strength and sensation loss over the past eight years. Over his lifetime, this musician acquired approximately 20 hardware synthesizers spanning modular, semi-modular, and keyboard formats to produce electronic music and live performances. Unfortunately, the impacts of progressive disability made it difficult for him to interact with his expansive and carefully curated instrument collection. Our goal was to create a tool that would allow this dedicated musician to continue his music practice, regardless of physical challenges [27].

To adapt to mobility loss, this musician used the following workarounds: 1) Apply a T-wrench for leverage on small knobs (limited to knobs with compatible geometries); 2) Drag a palm across control surfaces (imprecise, fatiguing); 3) Avoid instruments with densely packed or recessed controls; and 4) Rely on software-based synthesis (reducing tactile instrument interaction). Ultimately, these modifications reportedly led to frustration due to loss of expressive control and decreased motivation to engage with hardware instruments despite continued interest in this creative practice.

Once the custom knobs were generated on a 3D printer using ACCESS KNOB software, our musician required assistance to install an extremely large number of Access Knobs to upgrade his collection of synthesizers. Once installed, the musician was able to manipulate the knobs on his own, making the instruments more usable. As we tested the Access Knobs, we learned that application of the modified knobs requires assistance for the mobility-limited user. Pairing increased usability of these knobs with other adaptations has improved the user’s quality of life and reduced the burden of implementing imperfect workarounds, giving him more energy for other projects.

6 Discussion

Due to the difficulty of customizing the hardware for electronic and synthetic instruments, musicians have sadly set aside their instruments and their creative music practice once mobility limitations occur. However, music instruments can now be modified for accessibility using tools like ACCESS KNOB. Core features of this tool highlight user-driven, browser-based design that allows full customization of products to their motor profiles as well as easy and inexpensive production by individual users using 3D printers and laser cutters. This tool supports user agency and independence. The non-destructive slide-over sleeves preserve commercial hardware integrity, while tactile shape-coding bridges visual and motor accessibility. Releasing the tool as open-source lowers production costs to cents per knob, though it transfers quality control to user print profiles. Future work will extend the parametric engine to physical linear faders (mixer metaphor), drum pads/triggers, foot switches, and other custom interfaces while integrating color theory using LED’s and audio metering alongside haptic feedback; and will include occupational-therapy-guided clinical trials under an approved IRB protocol to evaluate long-term ergonomic utility. Utilizing UMD concepts, we will explore the “universal” capacity of ACCESS KNOB beyond this use case ported for DHH and neurodivergent users as well as

opportunities to support Universal Collaborative Artistic Practice. This work will be distilled into a curricular study for use in Accessibility and Special Education environments to evaluate efficacy while developing an original Accessible Music album.

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Anonymized for double-blind review.

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